INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION

PROVER, LIQUID VOLUMETRIC POSITIVE DISPLACEMENT METER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The meter prover is used to verify readings on displacement and turbine meters such as permanently installed meters used with truck fill stand installations or meters mounted on mobile refueling equipment such as the R-11 6000 gallon refueler and R-12 Hydrant Refueler. The prover is designed for the verification of positive displacement and turbine meters within the operational flow range of 30 GPM to 600 GPM with operating pressures not to exceed 150 psi. The meter prover consists of a trailer chassis and running gear that contains a master meter, suitable inlet and outlet piping and connections, discharge hoses and nozzles, control valves, check valve after meter, and pressure and 2 temperature gages. The master meter prover will be designed in accordance with API MPMS 4-1 and 4-5.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595/24052	Green, Semigloss
FED-STD-297	Rust proofing of Commercial (Non-tactical) Vehicles

COMMERCIAL ITEM DESCRIPTIONS

A-A-50696	Reels, Static Discharge Grounding, 50 and 75 feet Cable Lengths
A-A-59248	Meter, Volumetric, Positive Displacement, Liquid, Fuel, 600 GPM
A-A-59326	Coupling Half, Quick Disconnect, Cam locking Type

Comments, suggestions, or questions on this document should be addressed to WR-ALC/LER, 225 Ocmulgee Court, Robins AFB GA 31098-1647. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/online .

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624 Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST
MIL-PRF-85285 Coating, Polyurethane, High Solids
MIL-DTL-25524 Turbine Fuel, Aviation, Thermally Stable
MIL-DTL-83133 Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8), NATO F- 35, and JP-8+100 (NATO F-37)
MIL-DTL-83413 Connectors and Assemblies, Electrical, Aircraft Grounding, General Specifications
MIL-A-25896 Adapter, Pressure Fuel Serving, Nominal 2.5 Inch Diameter

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810 Environmental Engineering Considerations and Laboratory Tests
MS24484 Adapter, Pressure Fuel Servicing, Nominal 2.5 Inch Diameter

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil/online . or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASTM F 1347 Manually Operated Fueling Hose Reels, Standard Specification for

(ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org)

AMERICAN PETROLEUM INSTITUTE

API-STD-1529 Aviation Fueling Hose
API-RP2003 Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents
API-MPMS Chapter 4.5 Master Meter Provers
API-MPMS Chapter 4.9 Methods of Calibration for Displacement and Volumetric Tank Providers

(Copies are available from the American Petroleum Institute, 1220 L Street, NW, Washington DC 20005.)

SOCIETY AUTOMOTIVE ENGINEERS (SAE)

SAE-AS8090 Equipment, Towed Aerospace Ground, Mobility
SAE-AS5877 Nozzle, Pressure Fuel Servicing, Locking, Type D-1, D-2, D-3

(SAE documents may be obtained at http://www.sae.org/servlets/index or from SAE, Inc., 400 Commonwealth Drive, Warrendale PA 15096.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection.

3.2 Nozzle.

3.2.1 The nozzle connected to the hose on the hose reel shall meet or exceed the requirements of SAE-AS5877.

3.3 Hose reel. The hose reel shall meet or exceed the requirements of ASTM F 1347, Type 4 and shall contain minimum 50 feet of 2 1/2 inch of hose that conforms to API-BULL-1529.

3.4 Meter assembly. The volumetric positive displacement meter assembly shall be in accordance with A-A-59248. The meter shall include an automatic pre-set stop feature that will allow the operator to predetermine an amount of fuel to register and then automatically shut off the flow without causing a shock or hammer to the plumbing.

3.5 Single-point hose assembly. A single point hose assembly shall be provided that is connected to the swivel joint and that is 15 feet long and conforms to API-BULL-1529. A stowage facility shall be provided to coil the hose when not in use.

3.6 Swivel joint. A swivel joint shall serve as a connection between the single point hose assembly and the outlet side of the trailer system flow.

3.7 Pressure fuel servicing adapter. The adapter shall connect the meter assembly to the inlet side of trailer flow system. It shall have a 4-inch plug mounted a right angle to the adapter and a dust cap to prevent the entrance of foreign matter when the trailer is not in use.

3.8 Instrumentation. Instrumentation to monitor temperature and pressure through the operating range of the prover shall be provided and must be readable from a distance of 10 feet in average light conditions.

3.9 Static Grounding Reel. The reel shall be used to provide a contact between the trailer and the vehicle or facility. The assembly shall be in accordance with MIL-DTL-83413.

3.10 Valves. Valves of the appropriate type shall be used to close off the single point and hose reel delivery system when necessary.

3.11 Chassis and running gear. Chassis and running gear shall conform to SAE AS 8090.

3.12 Parking Brake. Parking brakes shall perform to SAE-AS8090.

3.13 Weight. The tester shall not exceed 1200 lbs. Gross dry weight.

3.14 Appearance. The unit and all components shall be painted in a tone down color of green, semigloss, Federal Standard 595, Color No. 24052, polyurethane coating MIL-PRF-85285.

3.15 Workmanship. The tester shall be manufactured meeting or exceeding standard commercial practices for the selection and installation of hardware, corrosion prevention practices and the ease of repair and servicing.

3.16 Performance. The tester shall be capable of withstanding exposure of the following conditions without adversely affecting the integrity, accuracy, and not cause any leakage.

a. Temperature ranges of -40°F to +130°F.

b. Fungus growth as encountered in tropical climates.

c. A relative humidity of 95 ±5% at a temperature of +130°F.

d. Sand and dust particles as may be encountered in desert areas.

e. Atmosphere containing salt-laden moisture.

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f. Being towed over semi-rough terrain and/or unimproved roads.

3.17 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable with similar parts with the same part number.

4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see para. 4.2)

b. Conformance inspection (see para. 4.3)

4.2 First article inspection. First article inspection, when required, shall be performed on one complete prover and shall consist of the following 4.4 examination, 4.5 methods of inspection, and 4.6 performance with a repeat of 4.6.1 at the completion of tests described by 4.6.3 through 4.6.8.

4.3 Conformance tests. Each prover shall be subjected to the following tests as described under 4.4., 4.5.2 and 4.6.1.

4.4 Examination. Each prover shall be examined for compliance with the requirements specified in 3.1 through 3.16.

4.5 Methods of inspection.

4.5.1 Testing apparatus. Apparatus used in conjunction with the testing specified herein shall be laboratory precision type and shall be calibrated at intervals properly spaced to assure laboratory accuracy.

4.5.2 Fuel. Unless otherwise specified, the fuel used during the testing specified shall conform to MIL-DTL-83133 and shall have a specific gravity of not less than 0.750 at 60°F.

4.6 Performance.

4.6.1 Hydrostatic test. The prover shall be filled and subjected to a pressure of 225 psi for a period of 10 minutes. All leaks encountered shall be repaired and the unit retested. No leakage shall be permitted.

4.6.2 Functional flow test. A test specimen meter shall be installed upstream of the prover in a pumping system capable of pumping at a rate of 600 gpm at 100 gpm increments. Each tested flow rate shall last for a period of ½ hour. The pre-set stop will be set at 5000 gal and 10000 gal incrementally at a 600 gpm rate. During this test the meter on the tester shall function in a satisfactory manner with no shock or hammer effect to the plumbing system.

4.6.3 High temperature test. The prover shall be subjected to high temperature in accordance with MIL-STD-810, method 501.5, procedure I, steps 1 through 4. Following the 48 hour exposure at 130° F, the prover shall be removed from the chamber and visually examined for deterioration. The prover shall show no deleterious effects from the exposure and shall complete the remainder .of the tests in a satisfactory manner.

4.6.4 Low Temperature test. The prover shall be subjected to low temperature in accordance with MIL-STD-810, method 502.5, procedure I, steps 1 through 6. Following the period of 24 hours after stabilization at -40° F, the prover shall show no evidence of failure and shall complete the remaining tests in a satisfactory manner.

4.6.5 Humidity test. The prover shall be subjected to humidity in accordance with MIL-STD-810, method 507.5, procedure I, steps 1 through 7. Following this test, the meter shall be visually examined for deterioration, evidence of swelling separation of components, and related types of failure.

4.6.6 Salt fog test. The prover shall be exposed to salt fog in accordance with MIL-STD-810, method 509.5, procedure I. During this test, the inlet and outlet of the meter shall be plugged to prevent internal contamination. The meter shall resist the effects of the test and shall show no evidence of failure or malfunction.

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4.6.7 Sand and dust test. The prover shall be exposed to sand and dust in accordance with MIL-STD-810, method 510.5, procedure I, steps 1 through 4. During this test, the inlet and outlet of the meter shall be plugged to prevent internal contamination. Upon completion of the test, it shall be demonstrated that the tester is completely operable.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging shall be as specified in the contract or order (see 6.4). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Military-Unique Rational. There are commercially available provers (master meters) that can prove fuel meters on fixed and mobile equipment; however, they must be able to verify total fuel delivery suitable for single point and auxiliary tank refueling of military aircraft style systems. It must also be capable of performing in the hostile environments of arctic and desert temperatures as well as maintaining its integrity on flightline surfaces as well as unimproved surfaces of forward operating locations.

6.2 Operator/Maintenance Manual. A detailed manual should be published to include, but is not limited to, master meter operations, operator maintenance, annual calibration requirements of master meter, and an illustrated parts breakdown.

6.3 Intended use. The type prover covered by this specification is intended to prove and verify readings of positive displacement and turbine meters installed in fixed facility refueling system, truck fillstands and mobile refueling equipment.

6.4 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Applicable levels of preservation, packaging, and packing required.
c. Data specified herein should be submitted in accordance with DD-Form 1423

6.5 Subject term (key word) listing.

Flightline
Fuel
Servicing
Totalizer
Trailer

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